

THEORETICAL AND EXPERIMENTAL WORK ON THE PHYSICS OF FAST REACTORS

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1. Introduction

The large fast reactor power stations now being proposed for electricity generation have posed a number of new physics problems. These are mainly because of the large size of the cores and their correspondingly low concentration of fissile atoms, and to the use of plutonium as a fuel in place of the U^{235} used in most existing fast reactors.

In order to provide the necessary physics data for the design of the prototype fast reactor (PFR) and for studies of future even larger reactors an extensive programme of both theoretical and experimental work has been necessary.

This paper reviews the calculation of some of the main physics properties of large fast reactors similar to the prototype fast reactor in the light of the integral measurements carried out in Zebra and other zero power reactors. From a comparison of theoretical predictions and experimental results it is possible to estimate the likely errors in predictions for the PFR and indicate where improvements in the data are desirable. A complementary paper⁽³¹⁾ considers some aspects of the expected physics behaviour of the PFR and some implications of the errors in the present predictions of the physics parameters.

In section 2 the methods of calculation used are outlined and the limitations due to heterogeneity effects and some approximations used in the calculations are discussed. Further limitations arise from the inaccuracies in the nuclear data fed into these calculations and the form of the data used and its probable accuracy are discussed in section 3. In the following sections the accuracy of calculations of the reactivity, important reaction rates, control rod worths and Doppler and sodium temperature coefficients are reviewed in the light of the information obtained from the integral experiments. This paper is

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primarily concerned with the recent results from the reactor Zebra⁽¹⁾ at Winfrith (Fig. 1). The results are supported and extended by measurements on the other zero power reactors and some references are also made to the

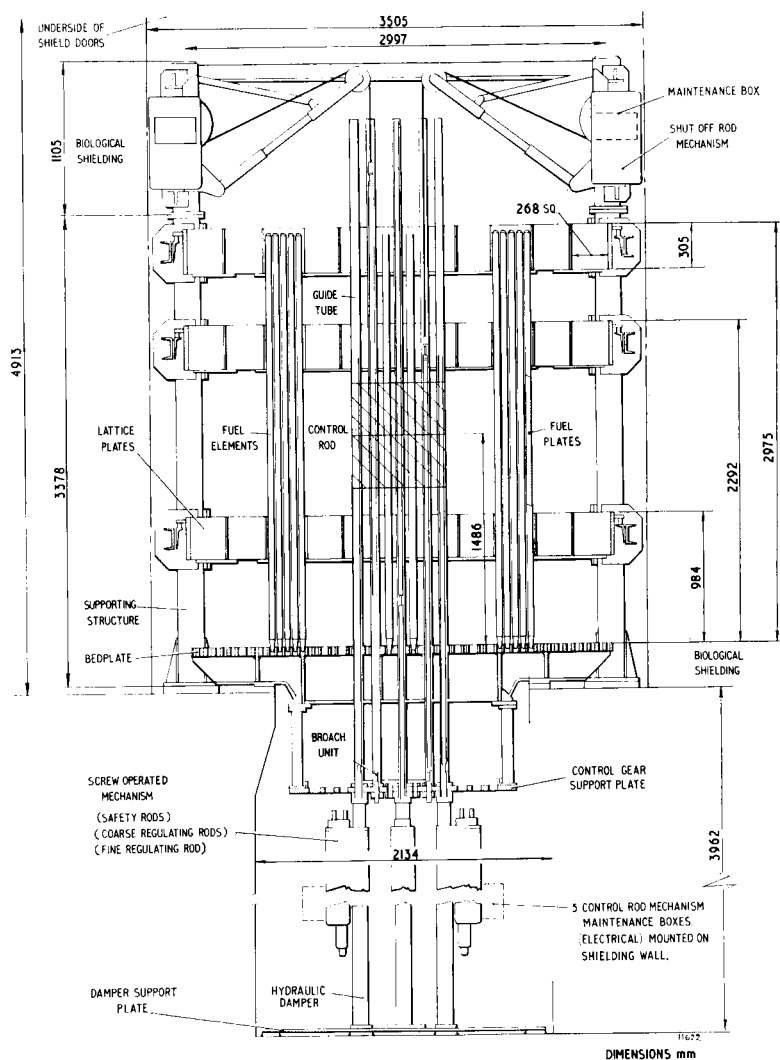


FIG. 1. Sectional view of Zebra.

work on the reactor Vera⁽²⁾ at Aldermaston and on the reactors ZPR III and VI^(3, 4) at Argonne National Laboratory in the U.S.A.

Six assemblies have been studied to date on Zebra and four of these (Nos. 1, 2, 3 and 6A) have had a uniform cylindrical core surrounded by a

uniform blanket. Assembly 4 also had a uniform core, but had three variants of the blanket. Assembly 5 was a multi-region system with a central heated test zone, used primarily for studying the Doppler effect. The core compositions are summarized in Table 1. The first assembly, U²³⁵ fuelled, was a repeat of ZPR III Assembly 11. The neutron spectrum in the core had a higher median fission energy than in a power reactor. The second assembly was also U²³⁵ fuelled and had graphite added to the core to make the neutron

TABLE 1. COMPOSITION OF CORES OF ZEBRA ASSEMBLIES (Atoms/cm³ × 10²¹)

	1	2	3	5J	6A
Pu ²³⁹	—	—	3.465	0.787	1.879
Pu ²⁴⁰	—	—	0.183	0.044	0.144
Pu ²⁴¹	—	—	0.016	—	0.016
U ²³⁵	4.655	2.523	0.229	0.019	0.046
U ²³⁸	34.33	15.55	31.56	4.368	6.353
Fe	4.124	3.856	4.559	9.155	4.525
Ni	0.424	0.397	0.321	1.211	0.443
Cr	1.077	1.008	0.818	2.457	1.270
Na	—	—	—	8.375	4.474
Al	—	—	—	—	2.504
O	—	—	—	10.96	—
C	—	37.95	—	—	29.59
Cu	—	—	4.574	—	0.829

spectrum approximate to that in a power reactor. Assembly 3 was similar to Assembly 1 but plutonium fuelled. Assembly 4 was derived from Assembly 2 by varying the blanket only. A 120° sector of the radial blanket of natural uranium was replaced by a four-region arrangement similar to that proposed for a power reactor. This was Assembly 4A. Assemblies 4B and 4C had two versions of axial blanket to study the effect of heterogeneity. Assembly 5 had fourteen variants of the test zone. A representative composition, for Assembly 5J, is included in Table 1. Assembly 6 had a plutonium fuelled core containing some graphite to obtain a neutron spectrum similar to a power reactor yet with a small critical mass. It also contained sodium to allow some information on the sodium loss coefficient to be obtained.

2. Methods of Calculation for Fast Reactors

Most calculations of the neutron physics behaviour of the Zebra assemblies and of the PFR itself have been made using multi-group diffusion theory in one and two dimensions. In addition, certain aspects have also been calculated using multi-group S_n methods, and Woodcock *et al.* have made a number of studies using the GEM Monte Carlo code⁽⁵⁾ (which obtains its

nuclear data by interpolation in energy tables). More refined methods have also been used for calculating the Doppler coefficient, heterogeneity effects, and control rod worths. In the diffusion theory calculations, each region (core, blanket, and reflectors) is treated as homogeneous and the control rods are smeared through an annular region having a thickness equal to the diameter of the rods. Critical enrichments, Doppler, sodium, and thermal expansion coefficients, breeding gain, power distributions and changes of reactivity with burnup are calculated using a simplified model of this kind. Details of the model are given in a complementary paper.⁽³¹⁾

These calculations have a number of limitations, amongst them being the effects of heterogeneity and practical limitations in the number of mesh points that can be used in diffusion theory calculations.

The variation of the U^{238} fission rate across a cell was $\pm 10\%$ in Zebra 3 and the U^{238} capture rate varied by $\pm 25\%$ across a cell in Zebra 6. An equivalent homogeneous critical size for the zero power assemblies is obtained by measuring the change in reactivity when a region of the core is bunched (the cell dimensions being doubled) and by assuming the heterogeneity effect to be linear with cell size. This assumption has been shown to be reasonable except when resonance shielding effects are important.

Two aspects of heterogeneity have been studied in detail. The first is the effect of the variation of the group averaged flux through a lattice cell, and this has been studied using transport theory codes and a perturbation theory which uses collision probabilities.⁽⁶⁾ The second is the effect of the stronger resonance shielding in a heterogeneous composition, and this has been studied using the GENEX-SDR codes,⁽⁷⁾ and approximate equivalence relationships (relating heterogeneous to equivalent homogeneous compositions). The first effect usually leads to an increase in reactivity with increase in cell size because of the increased probability of fission neutrons making a collision in the fissile material before escaping from it. In the second case, heterogeneity increases the resonance shielding in the fissile and fertile isotopes, thus reducing the group averaged cross-sections. The predominant effect is usually a reduction in capture in the fertile isotope and an increase in the Doppler coefficient of the fissile isotope, but there is also a reduction in the effective scattering cross-section in the 3 keV sodium resonance. These effects are all calculated to be small for the lattice structure of a power reactor, the Doppler effect and the moderation component of the sodium coefficient changing by a few per cent, and the reactivity increasing by less than 0.1%.⁽⁸⁾ The variation of group averaged flux in a lattice cell contributed about 1% in reactivity.

The difference between the reactivity calculated using S_n methods and diffusion theory is about 1% or 2% for small assemblies like Zebra 1 and 3 but it is only about 0.5% for Zebra 2 and is estimated to be smaller for the PFR (about 0.1%). Reaction rate scans calculated using S_n and diffusion

theory agree to within about $\pm 1\%$ for a homogeneous model corresponding to the Zebra 4 blanket studies.

A direct comparison of finite difference and Monte Carlo methods has only been made for a few limited cases. Calculations for some few group, homogeneous reactor problems using STRAINT (S_n)⁽⁹⁾ and GEM gave reactivities which agreed within the accuracy of convergence of the Monte

TABLE 2. DEPENDENCE OF RESULTS ON THE CALCULATIONAL MODEL USED

Case	Two-dimensional cylinder		Infinite cylinder with axial buckling		
	A	B	C	D	E
Radial mesh points	38	19	38	19	19
Axial mesh points	32	16	Axial buckling A	Axial buckling A	Axial buckling B
Reactivity	0.998	1.003	1.000	1.001	1.001
CENTRAL FLUX					
Group 1	1.07%	1.05%	1.07%	1.05%	1.09%
Group 18 (1 keV)	1.08%	1.12%	1.06%	1.09%	1.03%
Doppler coefficient ($\delta K/^\circ\text{C} \times 10^{-6}$)					
U ²³⁸ component	-6.46	-6.36	-6.22	-6.12	-5.82
Pu ²³⁹ component	1.70	1.65	1.58	1.54	1.51
Net Doppler	-4.76	-4.71	-4.64	-4.58	-4.31
Sodium coefficient ($\delta K/^\circ\text{C} \times 10^{-6}$)					
Leakage	-8.63	-8.55			
Moderation	5.21	5.30	4.93	4.99	4.73
Capture	0.58	0.58	0.58	0.57	0.86

The enrichment was the same for cases A, B, C, and D.

For case E, the enrichment was 2.6% higher to make k the same as D with an axial buckling 10% higher.

The axial buckling, A, was chosen to make case C just critical.

Carlo calculations ($\pm 2\%$). For the other calculations made using GEM the difference between these results and the diffusion theory results are consistent with the known differences in cross-section data.

Calculation of control rod worths is a matter of some difficulty. When no special boundary conditions are used diffusion theory predicts worths about 30% higher than S_n methods. Reasonable agreement has, however, been obtained with the control rod measurements made in Zebra 2 by using first order perturbation theory together with empirical shielding factors. A

Monte Carlo perturbation method now being developed should give a more satisfactory treatment of off-central control rods of complex geometry.

Reaction rate distributions within a few centimetres of region boundaries may not be accurately calculated using present methods of group averaging, because the resonance shielding factors are computed for an infinite medium. The accuracy of the methods of calculating reaction rate distributions in the vicinity of control rods has also to be further investigated, although an empirical method based on measurements made in Zebra 2 appears to be satisfactory.

The dependence of the accuracy on the number of mesh points used in a diffusion theory calculation, and the accuracy with which properties can be predicted using a one-dimensional cylindrical model with an axial buckling, is shown in Table 2. The calculations are for a large sodium cooled fast reactor like the PFR and used the FD 1 nuclear data set.

3. Nuclear Data

The diffusion theory calculations used for fast reactors require nuclear data in many energy groups. The nuclear data measurements made in nuclear physics laboratories provide the raw data for the construction of such a multi-group data set, but a choice has to be made of the number of groups used and a neutron spectrum assumed within each group in order to perform the necessary averaging process.

The group cross-section set which has been used for most recent studies is the 33-group FD 2 set.⁽¹⁰⁾ This set has superseded the earlier set FD 1,⁽¹¹⁾ derived from the Yiftah, Okrent, and Moldauer⁽¹²⁾ set by the addition of low energy groups. The FD 2 set was designed for the study of the core properties of large sodium cooled fast reactors, the weighting spectrum which was used to derive most of the group constants being a typical spectrum of the core of such a reactor. There is one group covering neutron energies above 3.67 MeV together with thirty-two one-half lethargy width groups covering the range down to 0.414 eV. Most of the cross-sections were obtained from the U.K. Nuclear Data Library⁽¹³⁾ using the group averaging code GALAXY.⁽¹⁴⁾

These cross-sections are therefore based on nuclear data measurements, without modification to fit integral measurements. Resonance shielding factors for uranium and plutonium isotopes are calculated by the diffusion theory programme appropriately for each region of the reactor. The resonance shielding of the sodium cross-sections and the elastic moderation cross-sections for oxygen and steel were calculated for a typical large sodium cooled fast reactor using the code ELMOE, but these cross-sections are not modified to suit a particular composition.

For the calculation of core properties the first twenty-four groups (covering

the range down to 37.3 eV) are adequate, but for the calculation of some blanket reaction rates, when there is a reflector region between the core and the blanket, thermal data must be added. Reaction rates which have a significant contribution from below 0.414 eV have been studied with the one-dimensional S_n code GMS⁽¹⁶⁾ using a 40-group set. This set consists of FD 2 with seven groups added below 0.414 eV to treat the thermal region and was developed to study plutonium fuelled, hydrogen moderated, systems.

TABLE 3. EFFECT OF VARYING THE NUMBER OF DOWN-SCATTER GROUPS

No. of down-scatter groups	Reactivity	Per cent change in flux at 1 keV
10	0.9895	
8	0.9896	0.14
6	0.9899	0.92
4	0.9919	3.7

TABLE 4. EFFECT OF NUMBER OF LOW ENERGY GROUPS

No. of groups	Lower energy limit eV	Reactivity	Core breeding gain	Radial blanket breeding gain	Pu ²³⁹ fission rate in the blanket
24	37.3	1.0013	-0.34	0.24	0.55
33	0.414	0.9991	-0.33	0.24	0.54
35	Thermal	0.9987	-0.33	0.24	0.63

Diffusion theory calculations have also been made using a 35-group set which has two low energy groups (taken from the Hansen and Roach set) added to the FD 2 set. Upscatter is not included in this 35-group set because this would necessitate the specification of the full scatter matrix in the diffusion theory programme, as it is written at present, and this would exceed the storage capacity of the computer for most problems of interest. The computing time for a 40-group problem using GMS is long and so there is an incentive to make diffusion theory calculations. A simple treatment of the thermal region without upscatter is expected to be adequate but this question is being investigated further.

The FD 2 set contains ten down-scatter groups but six are sufficient for most purposes, as the calculations for the PFR given in Table 3 show.

The dependence of calculated properties on the lower energy of the group

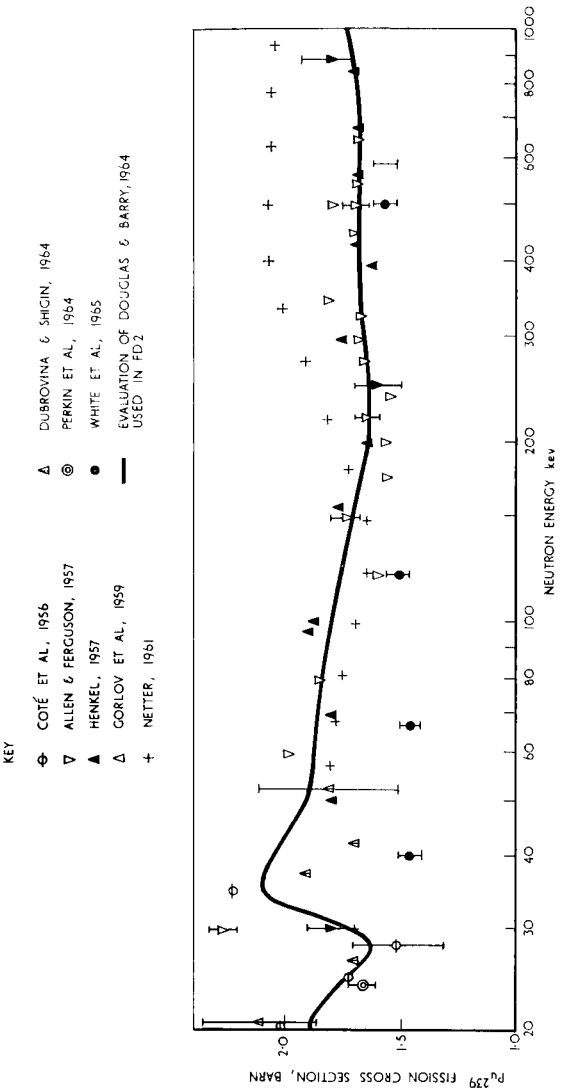


FIG. 2. Present and other measurements of the Pu^{239} fission cross-section.

structure has been examined by making one-dimensional cylindrical calculations with an axial buckling, and the results are given in Table 4.

The weighting spectrum used to derive the FD 2 set is inappropriate for blanket and reflector reaction rate calculations and a modified set has been produced to examine the importance of the correct choice of weighting spectrum for the calculation of these reaction rates.

The nuclear data used in the construction of these multi-group sets contain many uncertainties. As an illustration, the Pu^{239} fission cross-section measurements are presented in Fig. 2.⁽¹⁷⁾ The continuous line is that used in the FD 2 set⁽¹⁸⁾ and the black circles are the most recent measurements made at Aldermaston, which are about 30% below this line at 40 keV. These particular cross-section errors lead to important uncertainties in the calculated reactor properties, though these are not as great as might at first be thought since the data are least inaccurate in the energy region where most neutrons are found.

The effect of using the U^{235} and Pu^{239} fission cross-section values obtained in the recent measurements at Aldermaston and Harwell together with FD 2 data is to reduce the reactivity calculated for Zebra 2 by 1.2% from 0.992 to 0.980 and for Zebra 6A by 4% (from 1.010 to 0.970). In these two cases the calculated values are in poorer agreement with experiment, and this would apply generally to the calculations for dilute critical assemblies fuelled with plutonium.

Recent measurements at Harwell predict a fission spectrum for Pu^{239} which has a mean energy 7% higher than that for U^{235} .⁽¹⁹⁾ The Pu^{239} spectrum has 12% neutrons above 4 MeV compared with 10% for the U^{235} spectrum and consequently the fission spectrum averaged cross-sections for reactions with high energy thresholds are higher for a Pu^{239} fission spectrum. The effect of replacing the U^{235} fission spectrum by the Pu^{239} fission spectrum in PFR calculations is to increase reactivity by 1%.

4. Reactivity Calculations for Large Plutonium Fuelled Reactors

The designer of a power fast reactor will need an accurate prediction of the required plutonium content of each subassembly. It would be desirable to predict the required enrichment within $\pm 1\%$. No useful purpose would be served by better accuracy than this since the manufacturer will be hard pressed to make subassemblies to this tolerance. The corresponding limits for k are ± 0.005 .

Although there are outstanding problems in explaining the reaction rate distributions in complicated blankets, the change in reactivity with blanket design is small. For example, on changing the blanket from the well understood natural uranium blanket to a complete 360° blanket as built on Assembly 4A, there would have been a change in k of 0.017 (by extrapolation from the

experimental 120° sector). Useful information on the accuracy of reactivity calculations can therefore be deduced from studies of a variety of core compositions even though the blanket compositions are very different from those in power reactors.

The information available at the time that Zebra was first made critical included critical masses for more than twenty systems on ZPR III with critical masses of U^{235} up to 500 kg. Davey^(20, 21) has published a table of k values for these systems calculated using a transport theory programme, DSN, and a modification of the YOM set. It can be shown that the k values are normally distributed about 1.000 with a standard deviation of 0.006. The same models of ZPR III have been recalculated using the transport theory programme, WDSN, and FD 2, yielding a mean of 1.011 and a

TABLE 5. CALCULATED VALUES OF THE EFFECTIVE REPRODUCTION FACTOR, k , USING FD 1 AND FD 2, FOR ZEBRA ASSEMBLIES

Assembly	k FD 1	k FD 2
1	1.001	0.987
2	1.005	0.992
3	1.026	0.995
6A	1.043*	1.010*

* Based on preliminary values of certain corrections to the experimental result.

standard deviation of 0.015. It may therefore be concluded that the k value of an unknown system of similar composition and volume fractions to ZPR III assemblies can be calculated with a standard deviation of about 0.01 in k . Furthermore, practical power reactors will differ from ZPR III assemblies in two important respects. They will be fuelled with Pu^{239} not U^{235} and their core spectra will contain so many epithermal neutrons that the energy breakdown of the YOM set will be inadequate for calculations. The FD 2 set, on the other hand, has been deliberately designed to take into account epithermal neutrons and the self-shielding of resonance reactions.

It was the aim of the work on the first six assemblies of Zebra to show whether the agreement between experiment and calculation was worse when Pu^{239} was the fuel and when the core spectrum was made to include substantial numbers of epithermal neutrons by the addition of graphite to the core mixture.

A comparison of calculated values of the effective reproduction factor, k , for the just critical assemblies obtained using FD 1 and FD 2 is given in Table 5. The core constituents of Assembly 2 are similar to those of Assembly 1 except that graphite is also present; the core constituents of Assembly 6A are similar to those of Assembly 3 with graphite and sodium added. The

addition of graphite (which increases substantially the number of epithermal neutrons) does not apparently alter the accuracy of criticality calculations. The same four assemblies may also be considered as two pairs of corresponding assemblies, one with U^{235} fuel and the other with Pu^{239} fuel. The hard spectrum pair are 1 and 3; the soft spectrum pair 2 and 6A. Both sets predict a higher k for a plutonium fuelled assembly than for a similar assembly with U^{235} fuel but the difference is more marked for FD 1. The newer set, FD 2, predicts the value of k well for plutonium fuelled systems but gives a slightly low value for U^{235} fuelled systems.

It was pointed out by Baker and Ross⁽²²⁾ that the critical mass for a particular reactor core was closely related to the central perturbation cross-section of the fuel. It can be shown that the higher values of k predicted for Pu^{239} fuelled systems when compared with corresponding U^{235} fuelled systems are consistent with the errors in the calculation of the ratio of the central perturbation cross-sections of Pu^{239} and U^{235} measured in the assemblies in Table 6.

It may therefore be concluded that the change of fuel from U^{235} to Pu^{239} and the presence of substantial numbers of epithermal neutrons does not greatly worsen the agreement between calculated and measured values of reactivity.

In the case of the PFR verification of the critical enrichments will be obtained from the mockup of the reactor which it is proposed to build as Zebra Assembly 7.

5. Reaction Rates

Reaction rate measurements have been made in Zebra assemblies, with counters or foils, for fission in uranium and plutonium isotopes and capture in U^{238} . Checks of theory against experiment for Assemblies 1, 2, and 3 give an assessment of the accuracy of calculations for a fast reactor with a non-moderating breeder blanket. The corresponding check for the complex blanket of Assembly 4A shows the additional problems that arise when moderator is deliberately included in the blanket.

Reaction rate scans along the central axes of Assemblies 2 and 3 are given in Figs. 3, 4, and 5 for U^{235} fission, U^{238} fission, and U^{238} capture, respectively. All are normalized to unit fission rate in U^{235} at the reactor centre. The power distribution in a power reactor is principally due to Pu^{239} with a small contribution due to U^{238} fission. The accuracy of predicting Pu^{239} scans is similar to that shown for U^{235} in Fig. 3. The power distribution in a fast reactor core (with a non-moderating blanket) should be calculable to similar accuracy to that for U^{235} fission, namely within 2%. After a reactor has been operating for some time, the power production in the blanket due to fission of U^{238} and due to fission of bred Pu^{239} are both important. It

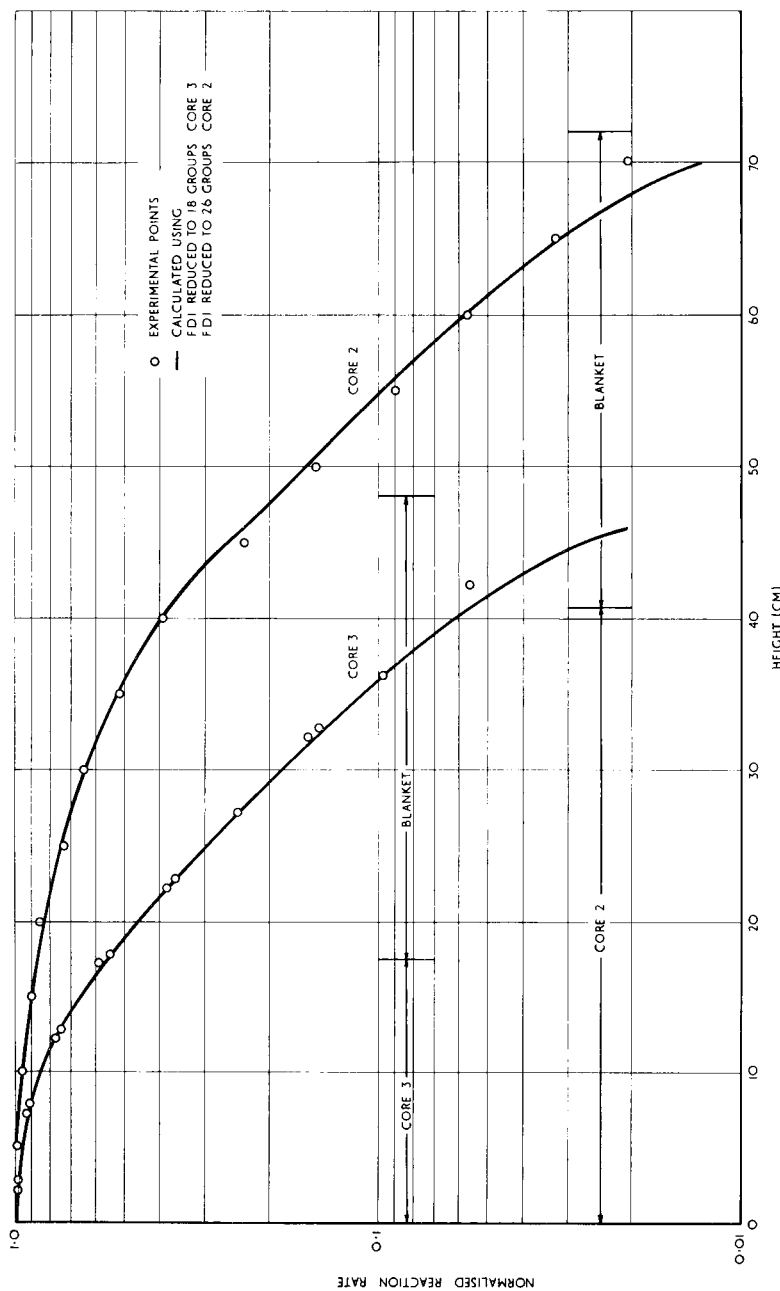


FIG. 3. U^{235} fission rates along the axis of Assemblies 2 and 3.

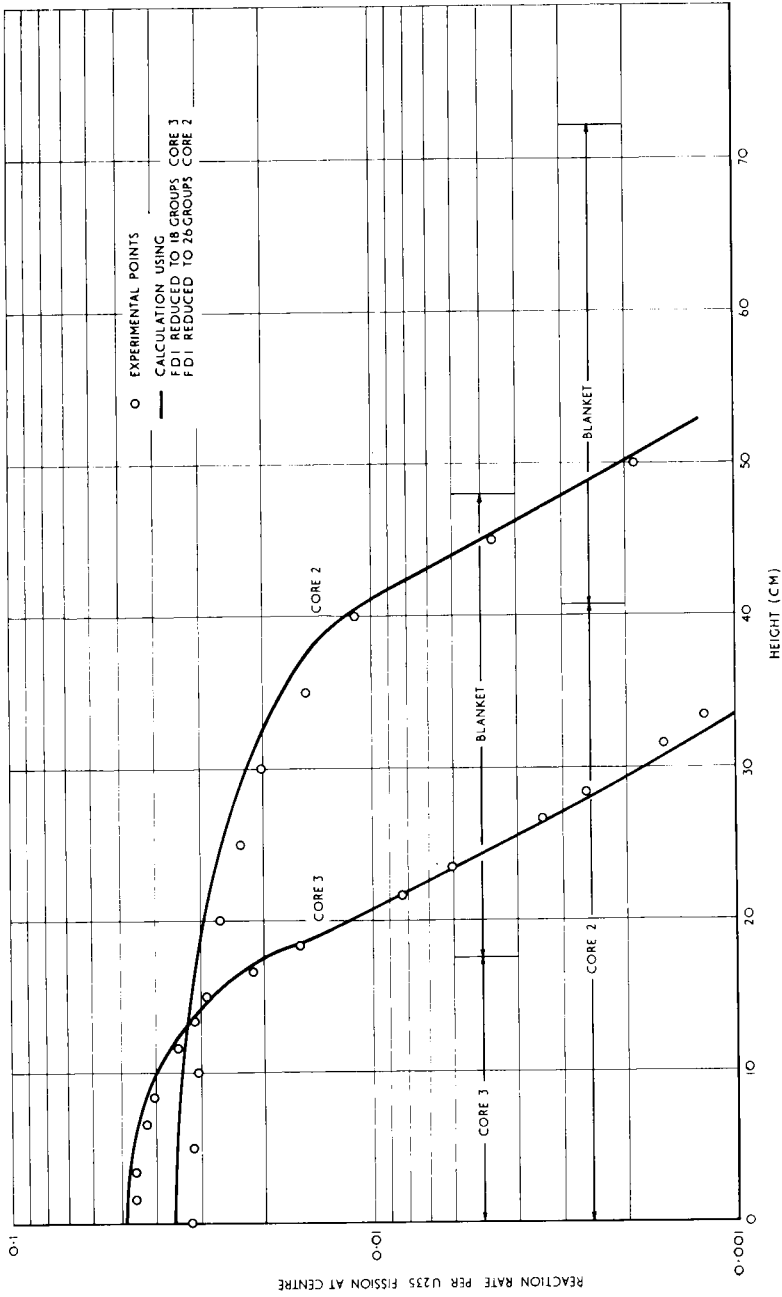


FIG. 4. U^{238} fission rates along the axis of Assemblies 2 and 3.

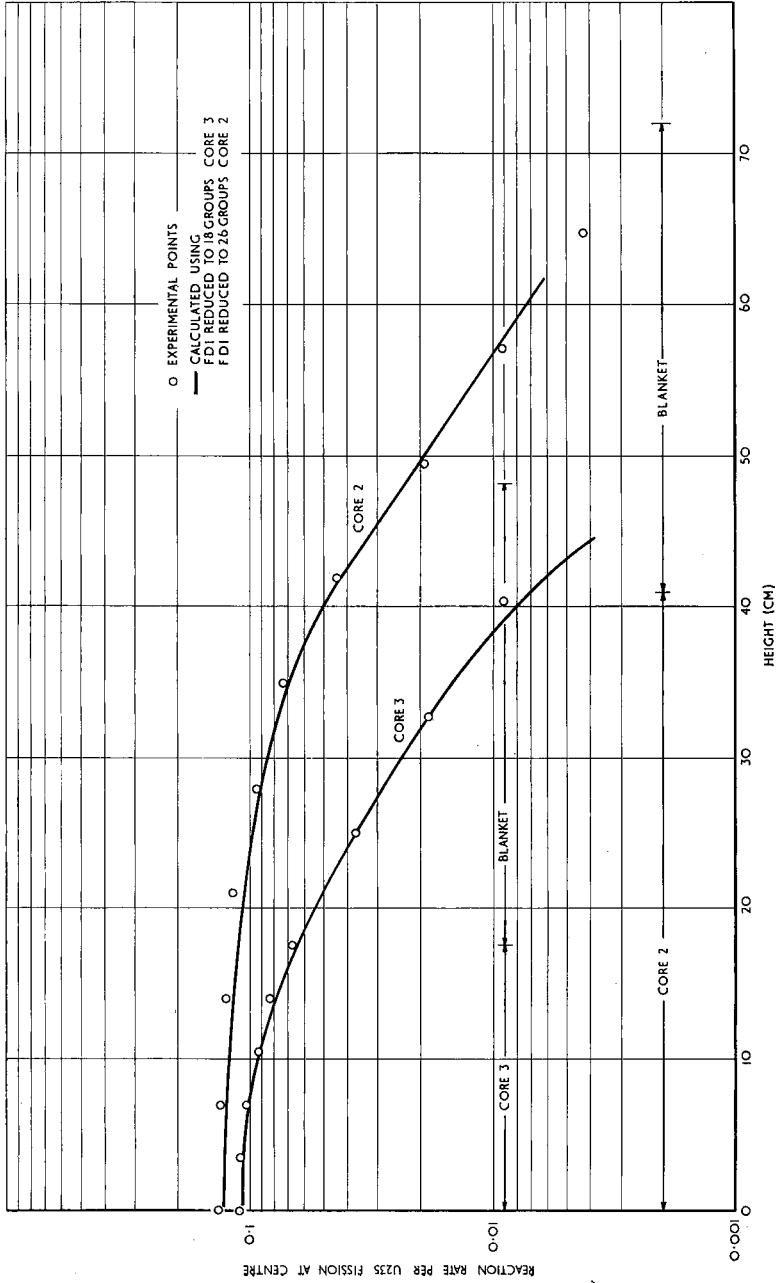


FIG. 5. U^{238} capture rates along the axis of Assemblies 2 and 3.

may be concluded, from Fig. 5, that the contribution due to Pu^{239} will be calculated using FD 1 within 10% but, from Fig. 4, that the contribution due to U^{238} may be overestimated by up to 15%.

It can be seen from Table 6 that there are errors in predicting relative fission rates in Zebra cores for different nuclides. These are mainly of importance as indicators of the detailed accuracy of calculations because it is not necessary to know the relative magnitudes of the contributors to the power distribution of the several nuclides. However, it is interesting to check the predictions of the change of fuel element power output with burnup. According to calculations by Okrent⁽²³⁾ for a particular reactor, a fuel element initially containing 23.3% Pu^{239} and 76.7% U^{238} would contain, after 10% burnup, 16.7% Pu^{239} ; 2.1% Pu^{240} ; 0.1% Pu^{241} ; 71.2% U^{238} and 10.0% fission products. According to the experimental reaction rate ratios in Table 6 the reduction in power output in a constant flux would be 24.0%; both FD 1 and FD 2 calculations give this figure within 0.1%. (The really difficult problem is in predicting the isotopic change with burnup. An experimental check can only be made in a power reactor.)

Some information on the accuracy of breeding gain calculations can be obtained from the data on U^{238} capture in Fig. 5 and Table 6. It can be seen that Pu^{239} production due to U^{238} capture in the cores and (non-moderating) blankets of Assemblies 2 and 3 is predicted within the experimental standard error of 5% using FD 1 data.

The measured Zebra 4A reaction rates give some indication that low energy reactions in the radial breeder are underestimated by calculation. However, there are still some uncertainties in the calculations relating the flux in the foils and counters to the flux in the region containing them. Table 7 shows the dimensions and compositions of the radial blanket of Zebra 4A and Fig. 6 shows the $\text{U}^{238}(n,f)$, Fig. 7 the $\text{U}^{235}(n,f)$ and Fig. 8 the $\text{U}^{238}(n,\gamma)$ reaction rates compared with calculations made using FD 2 data. The $\text{U}^{238}(n,\gamma)$ reaction rate is particularly sensitive to the resonance shielding assumed for the foil. In Fig. 8 the resonance shielding for the foils was assumed to be the same as that for the U^{238} in the breeder region. If a value corresponding to a foil in a non-resonant medium were to be assumed the calculated reaction rate would be doubled and would then agree with the measured value. An approximate calculation indicates a value midway between these two. However, the energy averaged flux may be higher in the foil than in the adjacent part of the breeder region, because of neutrons slowed down in the narrow non-absorbing region adjacent to the foil. The $\text{U}^{238}(n,f)$ reaction rate agrees with calculation in the radial breeder, but is higher than the calculated value beyond. The $\text{U}^{235}(n,f)$ reaction rate is underestimated by about 30% in the breeder. The effect of adding the two groups below 0.414 eV to the FD 2 set is to increase the U^{235} fission beyond the radial breeder, but the other reactions are not markedly affected.

TABLE 6. COMPARISON WITH THEORY OF SELECTED INTEGRAL MEASUREMENTS IN ZEBRA

Reaction cross-section ratio to $U^{235}(n,f)$	1	2	3	5J	6A
$Pu^{239}(n,f)$					
expt.	1.198 ± 0.014	0.987 ± 0.012	1.190 ± 0.014	0.904 ± 0.012	0.961 ± 0.013
FD 1	1.244	1.076	1.253	1.019	1.070
FD 2	1.193	0.990	1.191	0.884	0.948
$Pu^{240}(n,f)$					
expt.	0.355 ± 0.010	0.237 ± 0.003	0.373 ± 0.005	0.178 ± 0.002	0.252 ± 0.004
FD 1	0.373	0.248	0.405	0.184	0.270
FD 2	0.339	0.230	0.354	0.158	0.225
$U^{238}(n,f)$					
expt.	0.0409 ± 0.0007	0.0320 ± 0.0005	0.0461 ± 0.0008	0.0207 ± 0.0004	0.0364 ± 0.0007
FD 1	0.0420	0.0353	0.0480	0.0242	0.0428
FD 2	0.0370	0.0305	0.0403	0.0190	0.0329
$U^{238}(n,\gamma)$					
expt.	—	0.136 ± 0.007	0.111 ± 0.006	—	—
FD 1	0.111	0.129	0.109	—	—
FD 2	0.117	0.132	0.117	—	—
$B^{10}(n,\alpha)$					
expt.	—	1.58 ± 0.07	—	1.53 ± 0.07	1.63 ± 0.08
FD 1	0.49	1.07	0.48	1.20	—
FD 2	0.79	1.22	0.79	1.43	1.34

TABLE 6 (continued)

Perturbation cross-section ratio to U^{235}	1	2	3	5J	6A
Pu^{239}	expt. FD 1 FD 2	1.42 $\pm$ 0.02 1.55 1.48	1.61 $\pm$ 0.03 1.75 1.65	1.30 $\pm$ 0.02 1.47 1.29	1.44 $\pm$ 0.02 — 1.38
Pu^{240}	expt. FD 1 FD 2	0.27 $\pm$ 0.03 0.26 0.15	0.27 $\pm$ 0.05 0.53 0.37	— — —	— — —
U^{238}	expt. FD 1 FD 2	-0.078 $\pm$ 0.004 -0.065 -0.081	-0.050 $\pm$ 0.002 -0.045 -0.058	-0.073 $\pm$ 0.002 -0.075 -0.074	-0.062 $\pm$ 0.002 — -0.070
Ta	expt. FD 1 FD 2	-0.337 $\pm$ 0.007 -0.205 -0.260	-0.152 $\pm$ 0.007 -0.133 -0.137	-0.352 $\pm$ 0.012 -0.207 -0.238	-0.431 $\pm$ 0.002 — -0.261
Stainless steel $\times 10^3$	expt. FD 1 FD 2	-9.2 $\pm$ 0.9 -10.1 -8.4	— — —	-8.7 $\pm$ 0.4 -5.8 -5.5	-12.4 $\pm$ 0.4 — -7.2
Al $\times 10^3$	expt. FD 1 FD 2	-4 $\pm$ 1 -4 -4	-17 $\pm$ 2 -8 -11	-7 $\pm$ 1 -3 -4	-16 $\pm$ 1 — —
Na $\times 10^3$	expt. FD 1 FD 2	2.2 $\pm$ 1.8 1.8 1.3	-12.7 $\pm$ 1.5 -7.3 -11.6	-2.9 $\pm$ 1.2 -1.7 -3.0	-0.9 $\pm$ 0.4 — -1.7
C $\times 10^3$	expt. FD 1 FD 2	4 $\pm$ 1 1 4	-20 $\pm$ 1 -8 -15	— — —	2.1 $\pm$ 0.4 — 0.1
B ¹⁰	expt. FD 1 FD 2	-1.37 $\pm$ 0.04 -0.78 -1.02	-0.53 $\pm$ 0.02 -0.39 -0.45	-1.34 $\pm$ 0.03 -0.91 -1.09	-1.52 $\pm$ 0.08 — -1.14

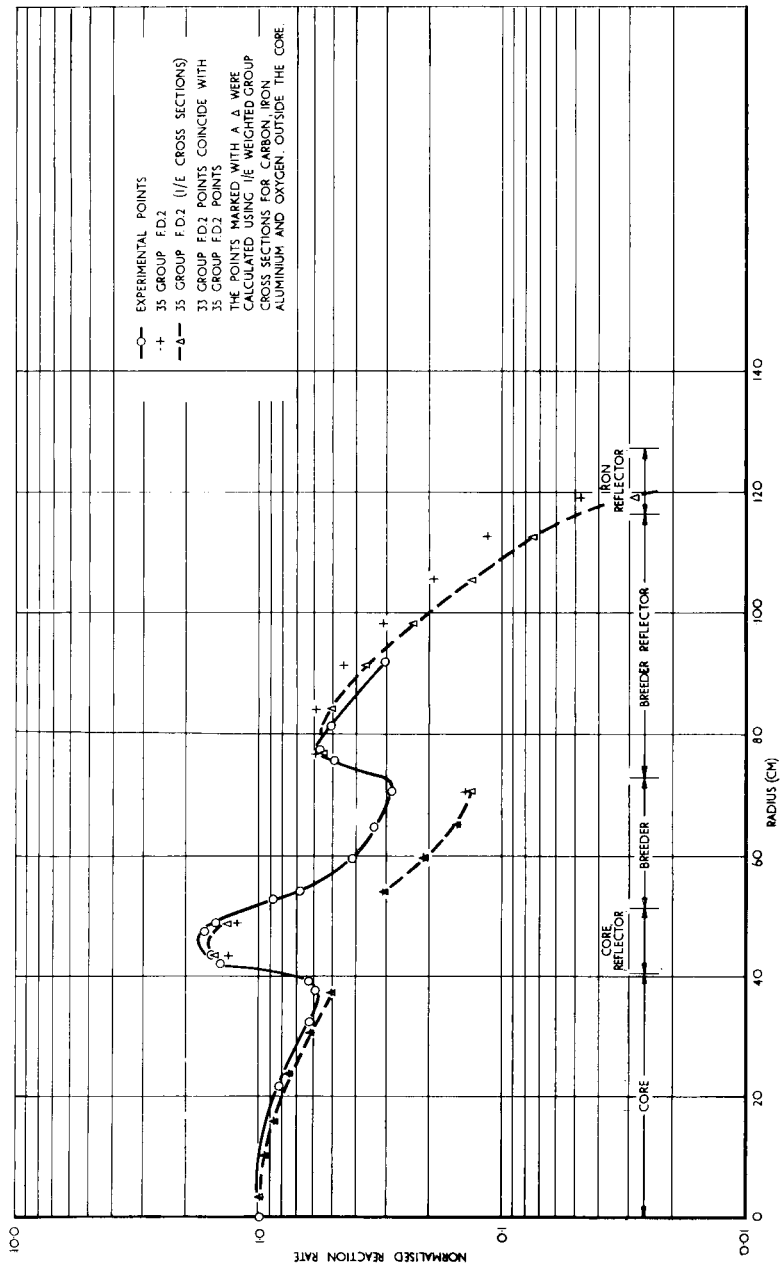


Fig. 6. U^{238} capture rates radially across the centre plane of assembly 4A.

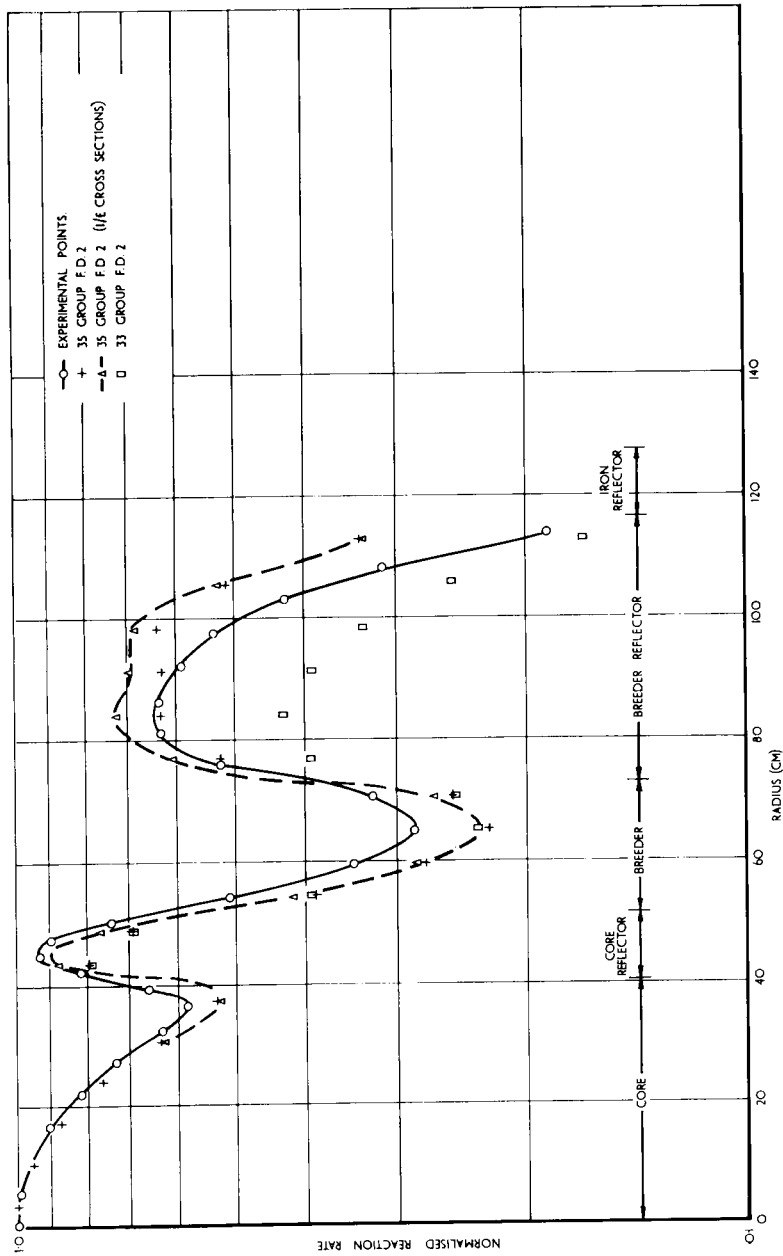


Fig. 7. U^{235} fission rate radially across the centre plane of assembly 4A.

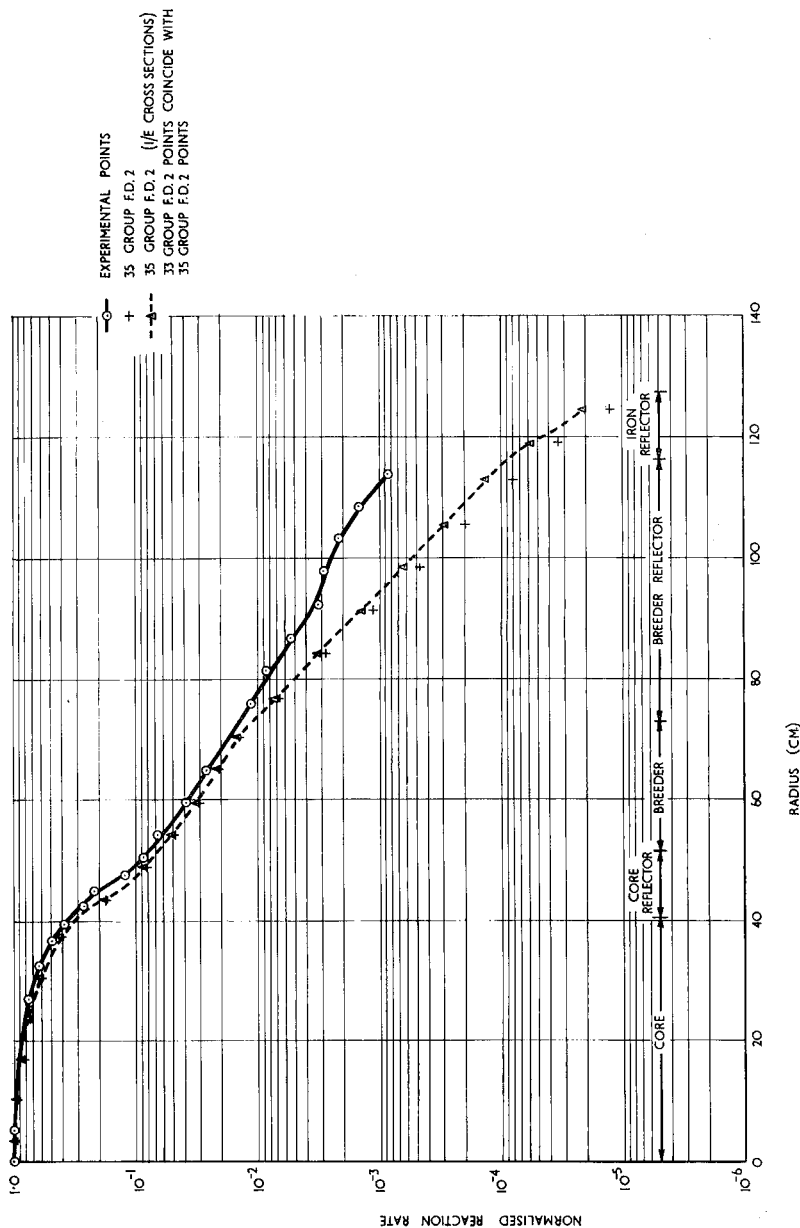


FIG. 8. U^{238} fission rates radially across the centre plane of assemble 4A.

Further work is in progress in an attempt to resolve these discrepancies between measurement and calculation for moderating blankets.

TABLE 7. DIMENSIONS AND COMPOSITIONS OF THE RADIAL BLANKET OF ZEBRA 4A

Zone	Effective thickness (cm)	Composition (atoms/cm ³ × 10 ²¹)							
		U ²³⁵	U ²³⁸	C	Fe	Cr	Ni	Al	O
Core	40.67*	2.523	15.55	37.95	3.86	1.01	0.40	—	—
1	10.87	—	—	37.91	28.91	7.55	2.98	—	—
2	21.74	0.096	9.57	—	9.83	0.91	0.36	10.85	19.33
3	21.61	—	—	64.99	6.61	1.73	0.68	1.48	—
4	21.61	—	—	70.99	3.50	0.91	0.36	—	—
5	11.00	—	—	—	77.17	0.91	0.36	—	—

* Core radius.

6. Control Rod Calculations

The control rods in a large dilute power reactor can conveniently be absorbing rods containing B¹⁰ or tantalum. Prediction of the reactivity worth of such rods is important and some information on the accuracy of design calculations can be obtained by comparing the experimental and calculated results in Table 6. Perturbation results are given for tantalum; BF₃ counter and perturbation results are given for B¹⁰. The perturbation experiments were performed with small samples and the results extrapolated to zero sample size.

At present, discrepancies between theory and calculation are large, as, for example, in Zebra Assembly 5J. For that assembly, calculations with FD 2 underestimated the worths of B¹⁰ and Ta by 20% and 30%, respectively. Even greater errors were observed with FD 1. These discrepancies are attributed to too few neutrons in the low energy tail of the calculated spectrum.

The measured reactivity effect per unit mass for a large absorber is smaller than the effect extrapolated to zero sample size (as given in Table 6). This occurs because of self-shielding within the absorber and local flux depression outside it. Some calculated and experimental values for long samples containing substantial amounts of B¹⁰ at the centre of Assembly 2 are given in Table 8. The calculated effect of the absorbers was obtained from diffusion theory calculations using the YOM set reduced to five groups. The change in reactivity was calculated in two stages, first for the substitution of core material for absorber, and then for removal of core material. The ratios of calculated to experimental values given in Table 8 can be compared with similar ratios of FD 1 results to experimental values for small samples given in Table 6. The discrepancies are smaller than in Table 6 because of compensating errors in the extrapolated value and the correction for self-shielding

and flux depression. It is possible that for the much thicker rods needed to control a power reactor the self-shielding and flux depression effects might over-compensate for the error in calculating small samples so that the calculated worth might be too great.

Activation measurements were also made with U^{235} foils close to one of the absorbers. The calculations were found to underestimate the flux depression in the neighbourhood of the absorber.

TABLE 8. COMPARISON OF MEASURED AND CALCULATED ABSORBING ROD* WORTHS IN ZEBRA 2

Absorber	Inside diameter (mm)	Outside diameter (mm)	Mass (g)	Measured worth ($10^{-4} dk/k$)	Calculated worth ($10^{-4} dk/k$)	<u>Calc.</u> Expt.
B ¹⁰	—	12.6	99.2	35.2 ± 0.7	27.2	0.77
B ¹⁰	—	19.0	186.8	64.6 ± 2.5	50.0	0.77
B ¹⁰	—	28.5	482.3	122 ± 4	109	0.89
B ¹⁰	15.9	28.5	269.0	88 ± 4	69.9	0.79
B ¹⁰	38.1	44.3	233.7	81 ± 4	70.7	0.87
Ta	—	6.0	399.6	3.1 ± 0.1	2.3	0.74
Ta	—	20.3†	5790	31.5 ± 0.9	2.83	0.90

* In each case, the length of the sample was the same as the core height, 83.4 cm.

† Square sample 20.3×20.3 .

At present, even though no calculation will give an accurate estimate of the central perturbation cross-section of B¹⁰, it is possible to obtain an estimate for a PFR design using a semi-empirical method and such a method is described in another paper.⁽³¹⁾

Transport theory calculations (S_4 approximation) for B¹⁰ control rods gave worths 20–30% smaller than diffusion theory calculations for rods having radii of 1–5 cm. The difference was smaller for smaller rods. The semi-empirical method referred to above is analogous to a collision probability theory calculation.

A Monte Carlo perturbation method for calculating control rod worths is being developed. This is designed to treat heterogeneous control rods which can be partially inserted or situated off the axis of the reactor. Only those neutrons which traverse the control rod volume are retracked.

Calculations by improved methods will only predict control rod worths satisfactorily when a cross-section set can be produced which will predict the core spectrum correctly. So long as it remains necessary to rely on semiempirical methods for calculating control rod worths, a substantial safety margin for error will have to be included in a power reactor design.

7. Sodium Temperature Coefficient

A reduction in the amount of sodium coolant present in a fast reactor core as a result of thermal expansion or loss of coolant from the whole or part of the core causes a change in reactivity. For convenience of calculation, this change is divided into three contributions:

- (i) a positive change due to the reduction in capture in sodium,
- (ii) a change, usually positive, due to the change in neutron spectrum, and
- (iii) a negative change due to the increased neutron leakage from the core.

In the PFR the sodium component of the temperature coefficient of reactivity is small. In normal operation the effect of coolant temperature changes is unimportant compared with the effect of fuel temperature changes, but reactivity changes due to sodium voiding may be important in safety studies.

The reactivity effect of sodium density changes is usually calculated using diffusion theory. Small density changes are treated using first order perturbation theory, which gives each contribution to within a few per cent of the exact diffusion theory calculation.⁽²⁴⁾ Calculations of the reactivity change resulting from the loss of sodium from a narrow spherical annulus in a spherical reactor made using diffusion theory and DSN gave results which agreed within a few per cent. There are still important uncertainties in the sodium cross-section data, for example, the resonance structure in the hundreds of keV region and the parameters of the 3 keV resonance. Uncertainties in the neutron importance below 100 keV caused, for example, by uncertainties in the Pu^{239} fission and capture cross-sections also have an important effect on the calculated sodium reactivity worth. Because of these uncertainties, measurements of sodium worth in zero power reactors have a special importance. However, there are indications that the heterogeneity of the composition of these reactors has a significant effect on sodium reactivity worth, and, at present, this complicates the interpretation of these results.

Information on the accuracy of calculations of the first two contributions above has been obtained from the central perturbation measurements with sodium given in Table 6. It can be seen that the experimental results agree with calculations using FD 2 within the accuracy of the measurements, which corresponds to a standard deviation of about 3 mb in the perturbation cross-section of sodium.

The reactivity effect of removing sodium from different regions of ZPR VI Assembly 2 has been measured.⁽²⁵⁾ The results have been analysed using FD 2. It was found that there was no systematic error in the calculation of the leakage component but an uncertainty of 10% cannot be excluded. Distributed sodium worth measurements are being made on Zebra 6 to study the leakage component.

A standard error of 3 mb in the central perturbation cross-section of sodium corresponds to an error of 0.006% in k in the effect of loss of sodium from the central subassembly and 0.3% from the whole core. A similar error would arise in the whole core effect due to a 10% error in the leakage component. The uncertainty in the reactivity worth of sodium in a large fast reactor with fresh fuel is therefore taken to be $\pm 0.5\%$ (equivalent to an uncertainty of $1.5 \times 10^{-6}/^{\circ}\text{C}$ in the coolant contribution to the temperature coefficient). The worth of sodium in a core corresponding to the average burnup condition with control rods partially inserted is calculated to be more negative by 1% in k . This arises from the increase in low energy capture due to fission products, control rod absorption, and the build-up of Pu^{240} (which has more threshold fission and more low energy capture than U^{238}). This difference must be regarded as uncertain to within about $\pm 30\%$ of itself, that is the decrease is $(1.0 \pm 0.3\%)$.

8. Doppler Temperature Coefficient

The fuel temperature coefficient (Doppler effect) is important for the assessment of the normal dynamic behaviour of a fast reactor and also for safety studies. Since the Doppler effect will be the principal contributor to the temperature and power coefficient of a large fast reactor it is necessary to know its value in order to design the reactor control system and to specify the required control rod worths. It has also been shown that the energy release following a nuclear excursion would be much reduced by the Doppler effect. This requires a knowledge of the Doppler effect to near the vaporization temperature of oxide fuel (about 2000°C).

The Doppler coefficient calculations made for the PFR have used diffusion theory and mostly first order perturbation theory, although the temperature dependence has been calculated by making diffusion theory reactivity calculations for a range of temperatures. The temperature-dependent group cross-sections for U^{238} and Pu^{239} were derived using resonance parameters and the code ERIC II⁽²⁶⁾ in the isolated resonance region, and using Nicholson's Method A⁽²⁷⁾ for the strongly overlapping resonance region. The PFR composition is treated as homogeneous in these calculations. The results have been compared with calculations made using the codes GENEX-SDR, which make no simplifying assumptions about the shape of the flux through resonances and energy groups and treat the heterogeneity of the composition by collision probability methods. These calculations have shown the effect of the heterogeneity of the PFR to be small and have given results in reasonable agreement with the simpler methods. The uncertainties in the U^{238} resonance data below about 10 keV are estimated to lead to uncertainties in the U^{238} component of the Doppler coefficient of less than about $\pm 10\%$ but the uncertainties in the low energy flux and importance spectra are greater

than this. The uncertainties in the low energy Pu^{239} cross-section data and resonance structure lead to much larger percentage uncertainties in this component of the Doppler coefficient because of uncertainties in the fission width distribution and the capture cross-section. An approximate upper limit (positive) to the coefficient can be estimated by assuming a constant

TABLE 9. COMPARISON OF MEASURED AND CALCULATED DOPPLER EFFECT FOR ZEBRA 2

	$\times 10^{-5} dk/k \text{ per } ^\circ\text{C}$
Measured contribution to temperature coefficient due to Doppler effect	-3.0 ± 0.7
Calculated effect: Hansen and Roach	-2.8
FD 1	-2.3
FD 2	-3.4

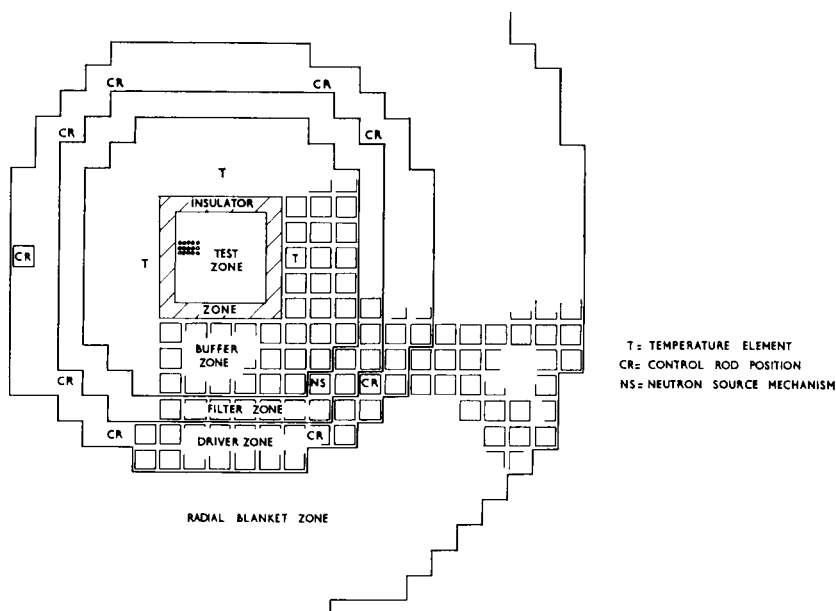


FIG. 9. Plan of Zebra Core 5A.

fission width and a low capture cross-section. This estimate is less than about one-quarter of the U^{238} coefficient. However, recent measurements indicate some very broad resonances in the total cross-section up to about 300 eV⁽²⁸⁾ (the present limit of resolved resonances) which are presumed to be predominantly in the fission cross-section. These may result in the Pu^{239} coefficient being negative (the fission cross-section being relatively insensitive to temperature changes).

TABLE 10. COMPOSITIONS OF THE ZONES OF THE DOPPLER LOOP EXPERIMENT, ASSEMBLY 5 (atoms/cm³ × 10²¹)

Region	Nuclide	5A	5B	5C	5D	5E	5H	5J	5K
Test	Pu ²³⁹	1.049	1.014	1.014	1.049	—	1.049	0.787	0.849
	Pu ²⁴⁰	0.059	0.057	0.057	0.059	—	0.059	0.044	0.048
	U ²³⁵	0.025	—	—	0.025	—	0.025	0.019	—
	U ²³⁸	5.82	—	—	5.82	—	5.82	4.37	—
	Fe	24.42	44.74	24.39	4.07	24.42	4.07	9.16	21.87
	Ni	3.23	3.23	0.54	0.54	3.23	0.54	1.21	2.89
	Cr	6.56	6.56	1.09	1.09	6.56	1.09	2.46	5.87
	Na	—	—	—	—	—	8.38	8.38	8.38
	O	14.61	2.14	2.14	14.61	—	14.61	10.96	1.80
	B	—	2.53	2.53	—	—	—	—	—
			As 5A		As 5C				As 5J
	Pu ²³⁹	1.144		1.082		As 5C	1.144	0.795	
	Pu ²⁴⁰	0.059		0.056			0.055	0.041	
Inner buffer	U ²³⁵	0.043		0.041			0.040	0.030	
	U ²³⁸	5.97		5.65			5.56	4.15	
	Fe	33.59		9.82			9.74	12.21	
	Na	—		—			6.68	7.54	
	Al	3.25		5.50			4.21	4.09	
	O	12.03		11.37			11.20	8.36	
	C	4.40		4.16			4.10	3.06	
	Cu	0.37		0.35			0.34	0.26	

Outer buffer	As 5A	1-075	As 5C	As 5C	As 5C	As 5J
Pu ²³⁹	1-133	—	—	—	—	0-801
Pu ²⁴⁰	0-060	—	—	—	—	0-042
U ²³⁵	0-041	0-039	—	—	—	0-029
U ²³⁸	5-62	5-34	—	—	—	4-03
Fe	30-34	8-49	—	—	—	12-31
Na	—	—	—	—	—	7-22
Al	3-06	4-86	—	—	—	3-92
O	11-33	10-75	—	—	—	8-01
C	4-14	3-93	—	—	—	2-93
Cu	1-49	1-42	—	—	—	1-06
Driver						
Pu ²³⁹	—	—	—	—	—	0-160
Pu ²⁴⁰	—	—	—	—	—	0-008
U ²³⁵	4-312	4-328	4-302	4-324	4-358	4-152
U ²³⁸	2-39	2-39	2-39	2-39	2-46	2-34
Fe	4-30	4-40	4-27	4-34	4-44	4-27
Cr	0-76	0-76	0-76	0-76	0-76	0-78
Ni	0-34	0-34	0-34	0-34	0-34	0-35
C	55-91	56-23	55-73	56-16	56-03	56-91
Cu	—	—	—	—	—	0-21

A first estimate of the accuracy of Doppler effect calculations was obtained from measurements of the core temperature coefficient in Zebra 2. This was not a precise experiment because of the non-uniform temperature distribution and because of the magnitude of the corrections for expansions. Preliminary results have already been published.⁽²⁹⁾ Revised results are given in Table 9. It can be seen that even this rough experiment confirmed that there was a substantial Doppler effect. A similar experiment for Zebra 3 confirmed that, in that case, the Doppler effect was negligible.

Previous accurate experiments on other zero-power reactors had used a small heated sample inside an unheated reactor. In order to eliminate some ambiguities in the interpretation of these experiments a new type of experiment was set up in Zebra 5 to check the Doppler calculations, including the temperature dependence. The reactivity changes were observed when a substantial region of a core was heated and cooled by circulating nitrogen. A plan of the arrangement is shown in Fig. 9. Fuller details of this experiment have been given at the Argonne Conference in 1965.⁽³⁰⁾

Zones of several compositions were used, as given in Table 10. The test zone contained a square array of fuel pins (about 20 cm side). The heated test zone was surrounded in each case by an unheated buffer zone of similar composition. A more reactive driver zone made the system critical. The arrangement used has been variously described as a "mixed critical system" or "internal exponential experiment". The test zone contained about 2% of the fuel in a complete reactor core and the reactivity change on raising the temperature from 100°C to 500°C was about 5×10^{-4} in k . The results are compared with calculations using three cross-section sets in Table 11.

In general, this experiment confirmed that Doppler calculations for large sodium cooled fast reactors are soundly based. The detailed conclusions may be summarized as follows:

- (i) A large negative Doppler effect in U^{238} was observed as expected, but the smaller positive effect due to Pu^{239} was not observed. More recent nuclear data than that incorporated in the FD 2 set suggests that the Pu^{239} effect may be very small. The best assumption for design calculations for the PFR is probably zero.
- (ii) The measured Doppler coefficients for the mixed Pu^{239}/U^{238} assemblies, 5A, 5D, and 5J were about 20% more negative than calculated with the most recent set, FD 2. If, however, the Pu^{239} contributions in the calculations are set to zero, then agreement between calculation and experiment is good, as can be seen from Table 11. This close agreement is fortuitous since other experimental evidence suggests that the low energy neutron flux is underestimated by calculation. This is apparently offset by errors in calculation of the adjoint flux.
- (iii) Some of the experimental results require a negative Doppler effect

for steel for a full explanation. The alternative explanation would be an unobserved expansion effect in the experiment.

- (iv) The Doppler effect in systems where the effect of U^{238} was dominant was found to vary as the reciprocal of the absolute temperature over the range from room temperature to 550°C .

TABLE 11. COMPARISON OF RESULTS AND CALCULATIONS FOR THE DOPPLER LOOP
(in units of $10^{-7}/^{\circ}\text{C}$)

Assembly	Measured temperature coefficient at 170°C (uncorrected)	Corrected temperature coefficient (Doppler effect)	Calculated Doppler coefficients			
			Hansen and Roach	FD 1	FD 2	FD 2 (U ²³⁸ only)
Assemblies with mixed oxide (A) pins 16% PuO ₂ , 84% U ²³⁸						
A	-7.7±0.07	-8.1±0.3	-6.3	-3.5	-6.6	-8.3
D	-6.0±0.1	-5.7±0.4	-5.0	-2.8	-4.6	-5.8
H	-5.3±0.1	-7.5±0.6	-5.7	-4.3	-6.3	-8.7
J	-4.0±0.1	-6.1±0.5	-4.5	-3.7	-5.0	-6.5
Assemblies with PuO ₂ without U ²³⁸						
B	-0.9±0.1	-0.8±0.3	+1.5	+1.5	+2.8	0
C	-1.5±0.1	-0.8±0.4	+1.1	+1.0	+1.8	0
K	-0.3±0.05	-2.9±1.0	+1.4	—	+3.0	0
Assemblies with non-fissile pins only						
E	-1.4±0.05	-1.3±0.5	0	0	0	0
L	-1.5±0.1	-1.7±0.2	0	0	0	0
M	-1.1±0.1	-1.0±0.2	0	0	0	0
J(4)	+1.2±0.1	-0.5±0.5	0	0	0	0
O	-0.3±0.1	-0.3±0.3	0	0	0	0
Assemblies without pins						
F	-2.35±0.05	-0.8±0.6	0	0	0	0
G	-2.3±0.05	-0.9±0.6	0	0	0	0
I	-1.75±0.05	-0.4±0.3	0	0	0	0

As a result of the Doppler loop measurements the uncertainties in the Doppler effect for a large fast reactor in its initial state have been reduced to about $\pm 30\%$. Because of uncertainties due to the effect of higher plutonium isotopes, fission products, and partially inserted control rods, the error for a practical reactor in the equilibrium state is perhaps double this.

9. Spectrum Measurements

It can be seen from the previous sections that, in many cases, it is impossible at present to deduce sufficiently exact information for a reactor design by calculation, even when supported by integral measurements made on a series of simple assemblies. As the next stage in the design of a PFR it is proposed

to build a mockup. On this it will be possible to measure criticality, reaction rates, control rod worths, etc. This repeats the pattern of the past where the Dounreay fast reactor was studied as a mockup on Zeus and successively EBR I, EBR II, AFSR, Fermi, Rapsodie, and SEFOR were studied as mockups on ZPR III.

TABLE 12. COMPARISON OF MEASURED AND CALCULATED NEUTRON SPECTRA IN CORE 5J

Group	Lower energy limit (keV)	Measured spectrum	Calculated FD 1	Calculated FD 2
1	3668	0.020	0.016	0.015
2	2225	0.042	0.034	0.037
3	1350	0.051	0.061	0.060
4	825	0.073	0.090	0.072
5	498	0.131	0.136	0.136
6	302	0.151	0.144	0.135
7	183	0.146	0.165	0.159
8	111	0.138	0.146	0.146
9	67.4	0.144	0.117	0.137
10	40.9	0.102	0.091	0.102

It might be possible to improve the accuracy of calculation if accurate measurements of reactor spectra could be made and attempts are being made on Zebra and Vera to develop methods of achieving this. Some preliminary results for the centre of Assembly 5J are given in Table 12. The spectrum above 400 keV was measured with a Li6 spectrometer and that between 40 and 1000 keV by proportional counters. The measured and calculated spectra have been normalized to a total of unity for the neutron energy range 40 keV upwards. According to FD 2 calculations, 25% of the spectrum lies below this energy for the centre of Assembly 5J. The standard deviation of a calculated group flux from the measured value is 16% for FD 1 and 12% for FD 2.

10. Conclusions

It is apparent that sufficiently accurate predictions can now be made for all the crucial physics parameters of a reactor of the PFR type to allow the construction of the reactor to proceed. At the same time it is clear that basic nuclear data have required extensive supplementation from integral experiments and that even with this experimental support in many areas increased accuracy is desirable, both for the efficient operation of the reactor and the optimization of future reactor designs. In order to provide some of this data for the PFR it is proposed to build a detailed zero-power mockup of the reactor as Zebra Assembly 7. It is probable that further integral experiments and more accurate nuclear data will reduce the need for mockups for future reactors, but it is not at present possible to foresee their elimination.

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